

Figure 8.1 Henrietta Leavitt.

But it is the young Belgian priest who understands, already in 1927, the crucial consequence: if we see a stone flying up, it means that the stone was previously lower down and something has thrown it upwards. If we see the galaxies moving away and the universe expanding, it means that the galaxies were previously much closer and the universe was smaller: and something caused it to start expanding. The young Belgian priest suggests that the universe was originally extremely small and compressed, and started its expansion in a gigantic explosion. He calls this initial state the primordial atom. Today it is known as the Big Bang.

His name was Georges Lemaître. In French, this name sounds like *le maître* meaning ‘the master’, and few names are more appropriate for the man who first understood the existence of the Big Bang. But in spite of this name, Lemaître’s character was reserved; he avoided polemics, and never even claimed priority for the discovery of the expansion of the universe, which ended up being attributed to Hubble. Two episodes from his life illustrate his profound intelligence. The first involves Einstein, the second a pope.

As mentioned, Einstein was sceptical about the expansion of the universe. He had grown up thinking that the universe is fixed, and had not been able to accept the idea that this was not the case. Even the greatest make mistakes and are prey to preconceived ideas. Lemaître met Einstein and tried to dissuade him from his prejudicial view. Einstein resisted, going so far as to answer Lemaître: ‘Correct calculations, abominable physics.’ Later, Einstein was obliged to recognize that Lemaître was the one who was actually right. It doesn’t fall to everyone to disprove Einstein.